CCRC Analysis Sheet

Program/Subject Area:	Artificial Intelligence Professional Development/Alignment Project (AI PD/AP)
Course(s):	A request for twenty-four (24) new AI-related courses was received.
Request:	Please refer to the introduction, table, and detailed course descriptions below for the proposed new courses.
Request Submitted By:	Forsyth Technical Community College on behalf of the AI PD/AP Team
Coordinator Assigned:	Dr. Hilmi Lahoud

Background:
The 2025/2026 Artificial Intelligence Professional Development/Alignment Project (AI PD/AP) was started in September 2025 and it is scheduled to end on June 30, 2026. All fifty-eight (58) community colleges were invited to participate. Forty-three (43) community colleges participated in the development of the proposed new courses. Forsyth Technical Community College was selected to serve as the resource college for the project. The AI PD/AP was funded by the Carl D. Perkins Strengthening Career and Technical Education for the 21st Century Act with funding from the U.S. Department of Education.

The objectives of this professional development/alignment project align with the Propel NC business model and include the following:

- Strengthening alignment between educational programs and North Carolina's evolving labor market by offering high-demand, high-wage programs and create pathways to economic mobility for students across the state.
- Work with the North Carolina Community College System to build partnerships with AI industries, enabling NC community colleges to use sponsored industry resources in course offerings.
- Develop new AI-related courses and supplements to be integrated into new and existing courses across multiple subject areas.
- Provide faculty with professional development activities and opportunities to share best practices for offering AI-related courses.
- Work with other stakeholders, including the NC Department of Public Instruction (DPI), advisory boards, multiple program areas, and industry professionals, to explore possible collaborations and partnerships.

The implementation date for the proposed new courses is Fall 2026.

Please refer to the table and course details below for more information:

Proposed New Courses			
Course #	Current Course Title	Core	Action Description
BAS-231	SAS Analytics Using AI I	No	**New:** Covers the current aspects of using AI in business analytics. **Rationale:** Aligns with the latest trends of using AI.
BAS-241	SAS Analytics Using AI II	No	**New:** Covers the current aspects of using AI in business analytics. **Rationale:** Aligns with the latest trends of using AI.
CCT-165	Threat Intelligence Analysis	No	**New:** Covers the current aspects of using AI in cybercrime. **Rationale:** Aligns with the latest trends of using AI.
CIS-123	Cisco Workplace App. of AI (Cisco Workplace Applications of Artificial Intelligence)	No	**New:** Covers the current aspects of using AI in the workplace. **Rationale:** Aligns with the latest trends of using AI.
CIS-127	AI for Manufacturing	No	**New:** Covers the current aspects of using AI in manufacturing. **Rationale:** Aligns with the latest trends of using AI.
CIS-129	Microsoft AI Fund with Azure (Microsoft AI Fundamentals with Azure)	No	**New:** Covers the current aspects of using AI in IT and databases. **Rationale:** Aligns with the latest trends of using AI.
CIS-131	AI for Sustainability	No	**New:** Covers the current aspects of using AI in sustainability. **Rationale:** Aligns with the latest trends of using AI.
CIS-132	AI for Accessibility	No	**New:** Covers the current aspects of using AI in education and learning. **Rationale:** Aligns with the latest trends of using AI.
CIS-133	AI for Agriculture	No	**New:** Covers the current aspects of using AI in agriculture. **Rationale:** Aligns with the latest trends of using AI.
CIS-134	Applied Ethics for AI	No	**New:** Covers the current aspects of ethics as it relates to AI. **Rationale:** Aligns with the latest trends of using AI.
CIS-142	Google AI Essentials	No	**New:** Covers the current aspects of using AI in the workplace. **Rationale:** Aligns with the latest trends of using AI.
CIS-167	Intel Applied Math for AI (Intel Applied Mathematics for AI)	No	**New:** Covers the current aspects of using AI in mathematics. **Rationale:** Aligns with the latest trends of using AI.
CIS-176	Intel Data Annotation	No	**New:** Covers the current aspects of using AI in data annotations. **Rationale:** Aligns with the latest trends of using AI.
CIS-214	AI Models	No	**New:** Covers the current aspects of using AI in AI-based systems.

			Rationale: Aligns with the latest trends of using AI.
CIS-221	AI Prompting (Artificial Intelligence Prompting)	No	**New:** Covers the current aspects of using AI in developing prompts for AI-based systems. **Rationale:** Aligns with the latest trends of using AI.
CIS-230	Microsoft Azure Data Science	No	**New:** Covers the current aspects of using AI in computer science. **Rationale:** Aligns with the latest trends of using AI.
CIS-231	Copilot & Agent Admin Fund (Copilot and Agent Administration Fundamentals)	No	**New:** Covers the current aspects of using Copilot as an AI agent. **Rationale:** Aligns with the latest trends of using AI.
CIS-232	Red Hat AI Concepts (Red Hat Artificial Intelligence Concepts)	No	**New:** Covers the current aspects of using Red Hat AI technology. **Rationale:** Aligns with the latest trends of using AI.
CIS-233	AI Deployment and Support	No	**New:** Covers the current aspects of AI system deployment. **Rationale:** Aligns with the latest trends of using AI.
CSC-164	Cisco AI/Python Data Science (Cisco AI and Python Data Science)	No	**New:** Covers the current aspects of using AI in data science. **Rationale:** Aligns with the latest trends of using AI.
CSC-165	Intel Natural Language Proc (Intel Natural Language Processing)	No	**New:** Covers the current aspects of using natural language AI in computer science. **Rationale:** Aligns with the latest trends of using AI.
CSC-166	Intel AI for Computer Vision	No	**New:** Covers the current aspects of using computer vision in computer science. **Rationale:** Aligns with the latest trends of using AI.
CSC-268	Apple Platforms: AI + AR	No	**New:** Covers the current aspects of using AI and augmented technology on Apple platforms. **Rationale:** Aligns with the latest trends of using AI.
SEC-165	AI in Security Operations	No	**New:** Covers the current aspects of using AI in security operations. **Rationale:** Aligns with the latest trends of using AI.

2025-2026 Artificial Intelligence Professional Development/Alignment Project (AI PD/AP)
Proposed New Course Descriptions

Proposed New Course
BAS-231 SAS Analytics Using AI I
Class: 2 Lab: 2 Credit Hours: 3
Prerequisites: None
Corequisites: None
Proposed Course Description:
This course covers data collection, data validation, data analytics, machine learning, generative AI using SAS, responsible innovation, and trustworthy AI. Emphasis is placed on the business aspects of SAS AI. Upon completion, students should be able to apply SAS AI concepts in the workplace.

Proposed New Course
BAS-241 SAS Analytics Using AI II
Class: 2 Lab: 2 Credit Hours: 3
Prerequisites: None
Corequisites: None
Proposed Course Description:
This course covers the advanced concepts of data science such as machine learning, statistical analysis and predictive modeling using SAS. Emphasis is placed on advanced analytics using SAS AI tools. Upon completion, students should be able to apply advanced SAS AI concepts in the workplace.

Proposed New Course
CCT-165 Threat Intelligence Analysis
Class: 2 Lab: 2 Credit Hours: 3
Course prerequisites: None
Course core requisites: None
Proposed Course Description:
This course covers the threat intelligence lifecycle and how it utilizes AI to enhance cybersecurity. Topics include data collection, pattern recognition, adversary profiling, and analytical reporting to strengthen intelligence-driven cyber defense, intelligence requirements, indicator analysis, OSINT collection, adversary TTP modeling, and AI-assisted analysis. Upon completion, students should be able to improve detection accuracy and intelligence processing.

Proposed New Course

CIS-123 Cisco Workplace App. of AI (Cisco Workplace Applications of Artificial Intelligence) Class: 2 Lab: 2 Credit Hours: 3

Prerequisites: None
Corequisites: None
Proposed Course Description:
This course is designed to cover the use of AI for common workforce tasks. Emphasis is placed on ethics, using Large Language Models (LLM) and AI image generation tools for workforce tasks such as analyzing customer reviews, and common business document generation. Upon completion, students should be able to utilize AI to perform customer reviews, examine resumes and generate other business documents.

Proposed New Course

CIS-127 AI for Manufacturing

Class: 2 Lab: 2 Credit: 3

Prerequisites: None
Corequisites: None
Proposed Course Description:
This course introduces Artificial Intelligence (AI) concepts related to the manufacturing sector. Topics include AI applications in quality monitoring, predictive maintenance, demand forecasting, and project lifecycle management. Upon completion, students should be able to analyze manufacturing challenges, design and implement AI-driven solutions, and develop ethical AI strategies for manufacturing environments.

Proposed New Course

CIS-129 Microsoft AI Fund with Azure (Microsoft AI Fundamentals with Azure)

Class: 2 Lab: 2 Credit Hours: 3

Course prerequisite: None
Course co-requisite: None
Proposed Course Description:
This course introduces foundational Artificial Intelligence (AI) concepts using the Microsoft Azure AI Fundamentals. Emphasis is placed on exploring common AI engineering workloads, Azure AI services, and responsible AI principles while gaining hands-on experience with AI models. Upon completion, students should be able to apply AI skills, utilize Azure tools in a working environment.

Proposed New Course

CIS-131 AI for Sustainability

Class: 2 Lab: 2 Credit: 3

Course prerequisite: None
Course co-requisite: None
Proposed Course Description:
This course introduces how artificial intelligence can support sustainability by addressing environmental, social, and economic challenges through responsible design, analysis, and application of AI technologies aligned with global sustainability goals. Emphasis is placed on AI fundamentals, the United Nations Sustainable Development Goals (SDGs), case studies of AI for sustainability, ethical considerations, impact evaluation, and hands-on projects using both low-code/no-code and coding-based AI tools. Upon completion, students should be able to develop, present, and apply AI concepts for SDG solutions.

Proposed New Course

CIS-132 AI for Accessibility

Class: 2 Lab: 2 Credit: 3

Course prerequisite: None
Course co-requisite: None
Proposed Course Description:
This course introduces how artificial intelligence can enhance accessibility by addressing physical, cognitive, and digital barriers through inclusive, ethical, and goal-driven AI solutions aligned with global sustainability and inclusion efforts. Emphasis is placed on accessibility concepts, accessibility challenges and opportunities, AI fundamentals, SDGs and inclusion, ethical and responsible AI frameworks, AI tools for remediation and treatment, and applied planning for AI-driven accessibility solutions. Upon completion, students should be able to address real-world accessibility challenges.

Proposed New Course

CIS-133 AI for Agriculture

Class: 2 Lab: 2 Credit: 3

Course prerequisite: None
Course co-requisite: None
Proposed Course Description:
This course introduces how artificial intelligence is applied in agriculture to improve productivity, sustainability, and decision-making through data-driven technologies and responsible AI practices. Emphasis is placed on AI fundamentals for agriculture, industry use cases, agricultural data and models, ethical and responsible AI considerations, the AI project lifecycle, and hands-on applications using low-code/no-code tools. Upon completion, students should be able to apply the AI project cycle to agricultural problems, and design and present a functional agricultural AI-based project.

Proposed New Course

CIS-134 Applied Ethics for AI

Class: 2 Lab: 2 Credit: 3

Course prerequisite: None
Course co-requisite: None
Proposed Course Description:
This course introduces applied ethics for artificial intelligence, focusing on ethical theory, responsible AI principles, and practical strategies for identifying, evaluating, and mitigating ethical risks throughout the AI system lifecycle. Emphasis is placed on AI ethics foundations, ethical pitfalls, ethical frameworks, ethics-first AI project cycles, explainable AI, data protection, impact assessments, post-deployment mitigation strategies, and the design of an organizational code of AI ethics. Upon completion, students should be able to apply ethical frameworks, identify and mitigate ethical risks when using AI in the workplace.

Proposed New Course

CIS-142 Google AI Essentials

Class: 2 Lab: 2 Credit Hours: 3

Prerequisites: None
Corequisites: None
Proposed Course Description:
This course introduces students to the essential AI concepts, tools, templates, and artifacts used to support individuals and businesses using Google resources. Emphasis is placed on fundamental AI methodology including the ethical aspects of AI and the use of generative AI to perform business tasks. Upon completion, students should be able to perform general AI business tasks within a business environment.

Proposed New Course

CIS-167 Intel Applied Math for AI (Intel Applied Mathematics for AI)

Class: 2 Lab: 2 Credit: 3

Course prerequisite: None
Course co-requisite: None
Proposed Course Description:
This course introduces essential mathematical foundations for artificial intelligence, helping students understand how mathematical concepts support AI models, data analysis, and decision-making across real-world applications. Emphasis is placed on the role of mathematics in AI, applied statistics, linear algebra, probability, and practical examples demonstrating how these concepts support each stage of the AI project cycle. Upon completion, students should be able to apply foundational mathematics utilizing AI to analyze complex solutions and develop scientific models.

Proposed New Course

CIS-176 Intel Data Annotation Class: 2 Lab: 2 Credit: 3

Course prerequisite: None

Course co-requisite: None

Proposed Course Description:

This course introduces data annotation as a foundational component of artificial intelligence, providing students with an understanding of AI domains and the role of high-quality labeled data in building effective AI systems. Emphasis is placed on an introduction to AI and its domains, data literacy fundamentals, data collection methods, data ethics, and hands-on exploration of tools and techniques used for data annotation. Upon completion, students should be able to apply Intel data annotations to provide business solutions in the workplace.

Proposed New Course

CIS-214 AI Models Class: 2 Lab: 2 Credit: 3

Prerequisites: None

Corequisites: None

Proposed Cours Description:

This course provides a comprehensive introduction to artificial intelligence models and their deployment, with a focus on the effective and purposeful use of AI technologies. Emphasis is placed on the history, development, practical utilization, and machine learning to build foundational knowledge of the field by applying hands-on skills. Upon completion, students should be able to research, select, and effectively utilize appropriate AI models to address business needs.

Proposed New Course

CIS-221 AI Prompting (Artificial Intelligence Prompting)

Class: 2 Lab: 2 Credit: 3

Prerequisites: None

Corequisites: None

This course introduces students to artificial intelligence prompting. Topics include defining the prompt task, providing sufficient context, specifying the desired format, and setting the appropriate voice or persona going into core techniques like zero-shot, few-shot, and Chain-of-Thought (CoT) prompting. Upon completion, students should be able to design, refine, and test sophisticated prompts for various AI models to reliably achieve desired outcomes while understanding the ethical considerations.

Proposed New Course

CIS-230 Microsoft Azure Data Science Class: 2 Lab: 2 Credit: 3

Course prerequisite: None

Course co-requisite: None

Proposed Cours Description:

This course introduces foundational concepts of Artificial Intelligence (AI), including machine learning, natural language processing, computer vision, generative AI, and responsible AI principles. Emphasis is placed on ethical considerations, terminology, and industry-aligned concepts associated with the Microsoft Azure AI framework. Upon completion, students should be able to apply Microsoft Azure data science solutions in a working environment.

Proposed New Course

CIS-231 Copilot & Agent Admin Fund (Copilot and Agent Administration Fundamentals)
Class: 2 Lab: 2 Credit Hours: 3

Course prerequisite: None

Course co-requisite: None

Proposed Course Description:

This course introduces foundational concepts for administering AI copilots and agents within Microsoft platforms. Emphasis is placed on exploring configuration, governance, security, and responsible use of copilots and agents, focusing on administrative oversight rather than model development. Upon completion, students should be ready to utilize Copilot and AI agents to perform basic administration tasks.

Proposed New Course

CIS-232 Red Hat AI Concepts (Red Hat Artificial Intelligence Concepts)
Class: 2 Lab: 2 Credit Hours: 3

Course prerequisite: None

Course co-requisite: None

Proposed Course Description:

This course covers machine learning, generative AI, large language models, natural language processing, computer vision, and responsible AI practices using Red Hat's open-source technologies. Emphasis is placed on the enterprise applications and ethical integration of AI. Upon completion, students should be able to apply Red Hat AI concepts to technical projects and organizational strategies.

Proposed New Course

CIS-233 AI Deployment and Support

Class: 2 Lab: 2 Credit Hours: 3

Course prerequisite: None
Course co-requisite: None
Proposed Course Description:
This course introduces AI deployment concepts within an existing infrastructure. Emphasis is placed on the AI deployment agent support and maintaining data center operations and technical support. Upon completion, students should be able to deploy AI solutions and perform administration tasks.

Proposed New Course

CSC-164 Cisco AI/Python Data Science (Cisco AI and Python Data Science)

Class: 2 Lab: 3 Credit Hours: 3

Course prerequisite: None
Course co-requisite: None
Proposed Course Description:
This course introduces the basics of the Python language, working with datasets, and using industry-standard tools provided within Python libraries. Emphasis is placed on cleaning, analyzing, visualization, modeling, and storytelling of datasets. Upon completion, students should be able to utilize Python and AI technologies to provide programmatic solutions with real-world datasets.

Proposed New Course

CSC-165 Intel Natural Language Proc (Intel Natural Language Processing)

Class: 2 Lab: 2 Credit: 3

Course prerequisite: None
Course co-requisite: None
Proposed Course Description:
This course introduces fundamental concepts of natural language processing, focusing on how computers analyze, understand, and generate human language through text-based data and practical applications. Emphasis is placed on text processing fundamentals, NLP concepts and techniques, language representation, and the design and development of a language recognition application using applied NLP tools. Upon completion, students should be able to recognize, design, develop, and present core NLP concepts, preprocess and analyze text data, and apply NLP techniques as it relates to Intel.

Proposed New Course

CSC-166 Intel AI for Computer Vision

Class: 2 Lab: 2 Credit: 3

Course prerequisite: None
Course co-requisite: None
Proposed Course Description:
This course introduces artificial intelligence techniques for computer vision, enabling students to understand how machines interpret visual data and apply AI methods to analyze and extract information from images. Emphasis is placed on image processing fundamentals, OpenCV and Python libraries, computer vision workflows, image classification, object detection, and applied AI techniques for visual analysis. Upon completion, students should be able to preprocess and analyze images, apply AI-based computer vision techniques and develop basic computer vision applications as it relates to Intel technology.

Proposed New Course

CSC-268 Apple Platforms: AI + AR

Class: 2 Lab: 2 Credit Hours: 3

Prerequisites: None
Corequisites: None
Proposed Course Description:
This course introduces artificial intelligence and augmented reality development on Apple platforms. Emphasis is placed on on-device machine learning using Core ML and MLX, Apple Intelligence features, Private Cloud Compute, AR/3D development with RealityKit, ARKit building intelligent, privacy-preserving applications and deploying optimized models within the Apple ecosystem. Upon completion, students should be able to develop AI-enhanced applications, integrate AR features, utilize Apple's machine learning frameworks, and implement secure AI workflows across Apple devices.

Proposed New Course

SEC-165 AI in Security Operations

Class: 2 Lab: 2 Credit Hours: 3

Course prerequisite: None
Course co-requisite: None
Proposed Course Description:
This course covers the application of artificial intelligence in cybersecurity. Emphasis is placed on applying AI techniques to enhance security operations and integrating AI tools into organizational security practices. Upon completion, students should be able to use AI tools to automate and enhance security operations within workplace environments.